

Problem Statement : 1

A forest department has installed surveillance cameras at different watchtowers. The watchtowers are connected in the form of a **Binary Search Tree (BST)**, where each node stores a unique watchtower ID.

Every morning, the control room needs to identify the **K-th smallest watchtower ID** to schedule a routine inspection.

Given the root of the BST and an integer **K**, return the **K-th smallest watchtower ID**.

Input:

First line: **N** (number of nodes)

Second line: **N** integers representing the BST (insert in the given order)

Third line: **K**

Sample Input

```
7
5 3 7 2 4 6 8
4
```

Sample Output

```
4

  5
 / \
3   7
/\  /\
2 4 6 8
```

Problem Statement : 2

A city has installed smart transformers connected as a Binary Search Tree.

Each transformer stores the power generated at that location.

Due to maintenance requirements, engineers must find the **Lowest Common Ancestor (LCA)** of two transformers.

The LCA is the lowest node that has both transformers as descendants (a node can be its own ancestor).

Input

First line: **N**

Second line: BST insertion order

Third line: Two transformer IDs **P** and **Q**

Sample Input

7
20 10 30 5 15 25 40
5 15

Sample Output

